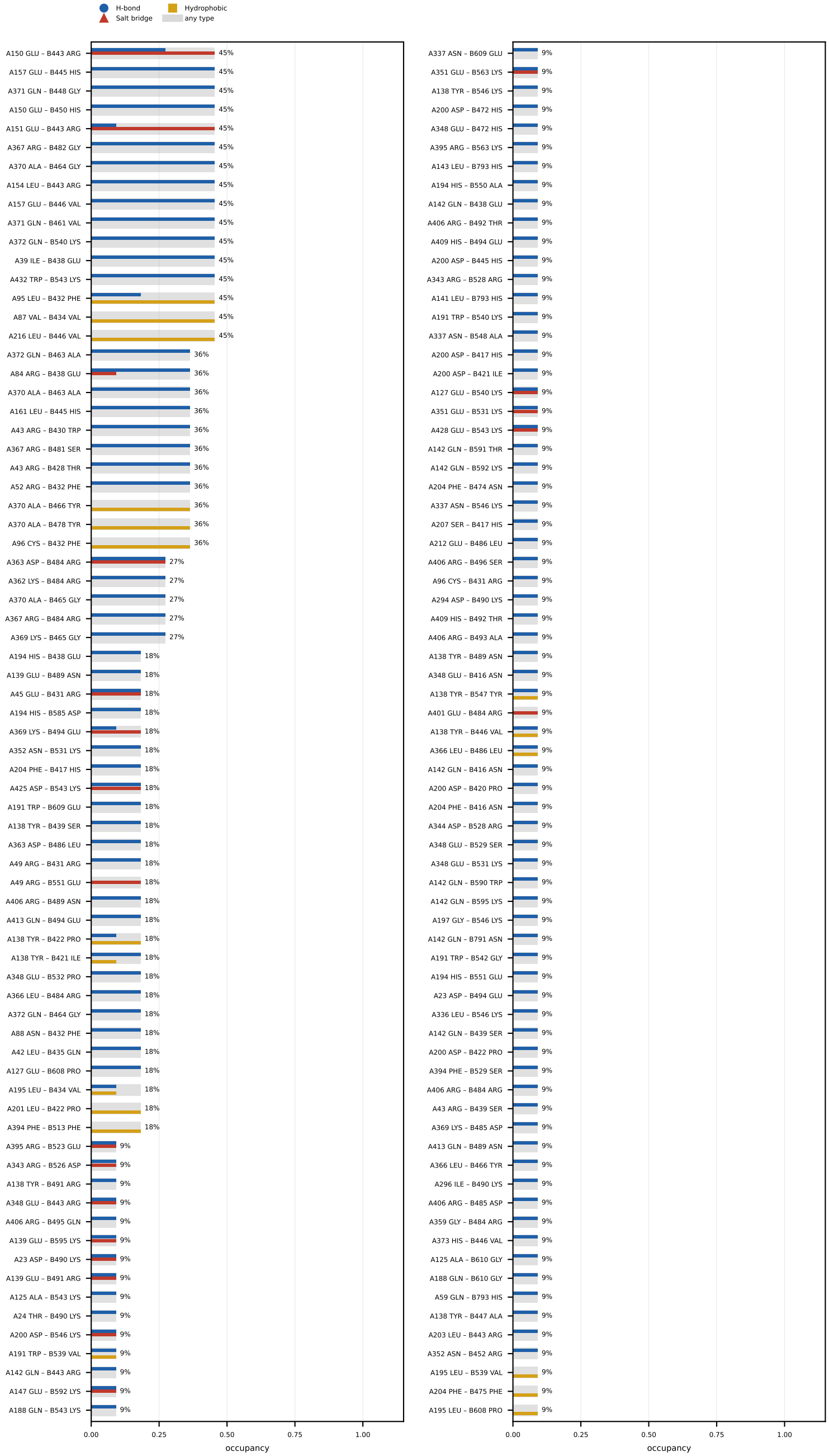E. All residue pairs (11 models, n=146)